Chapter 15

Body Fluids and Circulation

Every living cell has to be given nutrients, O2 and other essential substances, and the wastes it makes have to be taken away without pause, so tissues stay healthy only where movement in both directions is efficient.

How different animals solve it

- different groups of animals have evolved different methods for this transport
- simple organisms such as sponges and coelenterates drive water from their surroundings through their body cavities, letting the cells exchange these substances directly
- more complex organisms use special fluids held within their bodies to move such materials
- blood is the body fluid most higher organisms, humans among them, use for this purpose, and a second fluid, lymph, also carries certain substances

Blood, the tissue

- blood is a special connective tissue
- it consists of a fluid matrix, the plasma, and the formed elements

Plasma at a glance

- a straw coloured, viscous fluid making up nearly 55 per cent of the blood
- water accounts for 90 to 92 per cent of plasma and proteins for 6 to 8 per cent of it
- fibrinogen, globulins and albumins are the major proteins

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15.1 Plasma

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Neutrophil, eosinophil and basophil, the three granulocytes

Monocyte, with the B lymphocyte and the T lymphocyte, the agranulocytes

RBC, the biconcave red cell

Platelets, the small cell fragments

Figure 15.A Diagrammatic representation of formed elements in blood

15.1 Plasma

What the plasma proteins do

- fibrinogens are needed for the clotting or coagulation of blood
- globulins are primarily involved in the defense mechanisms of the body
- albumins help in osmotic balance

What else plasma carries

- small amounts of minerals such as Na+, Ca++, Mg++, HCO3–, Cl– and others
- glucose, amino acids and lipids, always in transit in the body
- the factors for coagulation or clotting, held in an inactive form
- plasma stripped of its clotting factors is called serum

15.2 Formed elements

Erythrocytes, the red blood cells

- the most abundant of all the cells in blood, at 5 millions to 5.5 millions of RBCs mm–3 in a healthy adult man on an average
- formed in the red bone marrow of adults, and in most mammals biconcave and without a nucleus
- haemoglobin, a red coloured iron containing complex protein, gives them their colour and their name, and a healthy individual carries 12 to 16 gms of it per 100 ml of blood
- these molecules matter greatly in transporting the respiratory gases, and an RBC lives on average 120 days before the spleen, the graveyard of RBCs, destroys it

Leucocytes, the white blood cells

- erythrocytes, leucocytes and platelets are collectively the formed elements, nearly 45 per cent of the blood
- leucocytes are colourless for want of haemoglobin, nucleated, and far fewer, averaging 6000 to 8000 mm–3 of blood
- generally short lived, they fall into two main categories: the granulocytes, that is neutrophils, eosinophils and basophils, and the agranulocytes, that is lymphocytes and monocytes

What each white cell does

- neutrophils are the most abundant of the total WBCs at 60 to 65 per cent, and basophils the least among them at 0.5 to 1 per cent
- neutrophils and monocytes, the latter 6 to 8 per cent, are phagocytic cells that destroy foreign organisms entering the body
- basophils secrete histamine, serotonin, heparin and the like, and take part in inflammatory reactions
- eosinophils, 2 to 3 per cent, resist infections

Eosinophils and lymphocytes

- eosinophils are also associated with allergic reactions
- lymphocytes, 20 to 25 per cent, come in two major types, the B and the T forms
- both B and T lymphocytes are responsible for the immune responses of the body

Platelets

- also called thrombocytes, they are cell fragments produced from megakaryocytes, special cells in the bone marrow
- blood normally contains 1,500,00 to 3,500,00 platelets mm–3
- they can release a variety of substances, most of which take part in the coagulation or clotting of blood
- a fall in their number brings on clotting disorders, and so an excessive loss of blood from the body

15.3 Blood groups

Two groupings used everywhere

- the blood of human beings differs in certain aspects although it appears to be similar
- various types of grouping of blood have been done, and two of them, the ABO and the Rh, are widely used all over the world

ABO grouping

- based on the presence or absence of two surface antigens on the RBCs, named A and B, an antigen being a chemical that can induce an immune response
- the plasma of different individuals likewise holds two natural antibodies, proteins produced in response to antigens
- the four groups A, B, AB and O differ in which antigens and which antibodies they carry

Matching a donor to a recipient

- during a transfusion any blood cannot be used; the blood of the donor has to be carefully matched with the blood of the recipient beforehand
- a mismatch brings on severe problems of clumping, the destruction of RBC
- group O blood can be donated to a person of any other group, so O group individuals are called universal donors
- persons of group AB can accept blood from AB as well as from the other groups, so they are called universal recipients

Table 15.A Blood Groups and Donor Compatibility

Blood Group	Antigens on RBCs	Antibodies in Plasma	Donor's Group
A	A	anti-B	A, O
B	B	anti-A	B, O
AB	A, B	nil	AB, A, B, O
O	nil	anti-A, B	O

15.4 Rh grouping

The Rh antigen

- another antigen, the Rh antigen, is similar to one present in Rhesus monkeys, which is where the name comes from
- it is observed on the surface of the RBCs of the majority of humans, nearly 80 per cent, who are called Rh positive (Rh+ve), while those in whom it is absent are Rh negative (Rh-ve)
- an Rh-ve person exposed to Rh+ve blood forms specific antibodies against the Rh antigens, so the Rh group too should be matched before transfusions

Incompatibility in pregnancy

- a special case of Rh mismatching has been observed between the Rh-ve blood of a pregnant mother and the Rh+ve blood of the foetus
- in the first pregnancy the placenta keeps the two bloods well separated, so the Rh antigens of the foetus do not get exposed to the blood of the mother
- during the delivery of the first child, however, small amounts of Rh+ve foetal blood may reach the maternal blood, and the mother then starts preparing antibodies against the Rh antigen in her blood

Erythroblastosis foetalis

- in her subsequent pregnancies the Rh antibodies of the Rh-ve mother can leak into the blood of the Rh+ve foetus and destroy the foetal RBCs
- this could be fatal to the foetus, or could leave the baby with severe anaemia and jaundice
- the condition is called erythroblastosis foetalis
- it can be avoided by administering anti-Rh antibodies to the mother immediately after the delivery of the first child

15.5 Coagulation of blood

Why a wound stops bleeding

- a cut does not go on bleeding for long, because blood coagulates or clots in response to an injury or trauma
- this is a mechanism that prevents excessive loss of blood from the body
- the dark reddish brown scum that forms at the site of a cut over a period of time is the clot or coagulam
- it is made mainly of a network of threads called fibrins, in which the dead and damaged formed elements of blood are trapped

The cascade behind the clot

- fibrins are formed when the enzyme thrombin converts the inactive fibrinogens of the plasma
- thrombins in turn are formed from prothrombin, another inactive substance present in the plasma
- an enzyme complex, thrombokinase, is required for that reaction, and is itself built by a series of linked enzymic reactions, the cascade process, drawing on a number of factors that lie inactive in the plasma
- injury stimulates the platelets to release certain factors that activate the mechanism, factors released by the tissues at the site of injury can also initiate it, and calcium ions play a very important role in clotting

15.6 Lymph, the tissue fluid

How tissue fluid forms

- as blood passes through the capillaries in tissues, some water together with many small water soluble substances moves out into the spaces between the cells of the tissues
- the larger proteins and most of the formed elements are left behind in the blood vessels
- the fluid released in this way is the interstitial fluid or tissue fluid, and it has the same mineral distribution as plasma
- exchange of nutrients, gases and the like between the blood and the cells always occurs through this fluid

The lymphatic system

- an elaborate network of vessels, the lymphatic system, collects this fluid and drains it back to the major veins
- the fluid present in the lymphatic system is called the lymph
- lymph is a colourless fluid containing specialised lymphocytes, which are responsible for the immune responses of the body
- it is also an important carrier for nutrients, hormones and the like, and fats are absorbed through lymph in the lacteals present in the intestinal villi

15.7 Circulatory pathways

Open and closed

- the circulatory patterns are of two types, open and closed
- the open system is present in arthropods and molluscs, where blood pumped by the heart passes through large vessels into open spaces or body cavities called sinuses
- annelids and chordates have a closed system, in which blood pumped by the heart always travels through a closed network of blood vessels
- the closed pattern is considered more advantageous, as the flow of fluid can be regulated more precisely

Chambers of the vertebrate heart

- all vertebrates possess a muscular chambered heart
- fishes have a 2-chambered heart with an atrium and a ventricle
- amphibians and the reptiles other than crocodiles have a 3-chambered heart with two atria and a single ventricle
- crocodiles, birds and mammals possess a 4-chambered heart with two atria and two ventricles

Single, incomplete double, double

- in fishes the heart pumps out deoxygenated blood, the gills oxygenate it and supply it to the body parts, from where deoxygenated blood is returned to the heart, which is single circulation
- in amphibians and reptiles the left atrium receives oxygenated blood from the gills, lungs or skin and the right atrium the deoxygenated blood from other body parts, but the two mix in the single ventricle, which pumps out mixed blood, so the double circulation is incomplete
- in birds and mammals the oxygenated and deoxygenated blood received by the left and right atria pass on to the ventricles of the same sides and are pumped out without any mixing, so two separate circulatory pathways are present and these animals have double circulation

Chambers, right and left atrium above, right and left ventricle below

Septa, inter-atrial between the atria, inter-ventricular between the ventricles

Atrio-ventricular septum, between the atrium and the ventricle of the same side

Valves, tricuspid on the right side, bicuspid or mitral on the left

Pericardium, the double walled bag around the whole organ

Figure 15.B Section of a human heart

15.8 The human heart

The heart in place

- the human circulatory system, also called the blood vascular system, consists of a muscular chambered heart, a network of closed branching blood vessels, and blood, the fluid which is circulated
- the heart is a mesodermally derived organ, situated in the thoracic cavity in between the two lungs and tilted slightly to the left
- it has the size of a clenched fist
- a double walled membranous bag, the pericardium, protects it and encloses the pericardial fluid

Four chambers, three septa

- the heart has four chambers, two relatively small upper chambers called atria and two larger lower chambers called ventricles
- a thin muscular wall, the inter-atrial septum, separates the right and the left atria
- a thick walled inter-ventricular septum separates the left and the right ventricles
- a thick fibrous tissue, the atrio-ventricular septum, separates the atrium and the ventricle of the same side

The valves between the chambers

- each of these septa is provided with an opening through which the two chambers of the same side are connected
- the opening between the right atrium and the right ventricle is guarded by a valve formed of three muscular flaps or cusps, the tricuspid valve
- a bicuspid or mitral valve guards the opening between the left atrium and the left ventricle

Semilunar valves and one way flow

- the openings of the right and the left ventricles into the pulmonary artery and the aorta respectively are provided with the semilunar valves
- the valves in the heart allow blood to flow in one direction only, from the atria to the ventricles and from the ventricles to the pulmonary artery or aorta
- they prevent any backward flow

Cardiac muscle and nodal tissue

- the entire heart is made of cardiac muscles, and the walls of the ventricles are much thicker than those of the atria
- a specialised cardiac musculature called the nodal tissue is also distributed in the heart
- a patch of it in the right upper corner of the right atrium is the sino-atrial node (SAN)
- another mass of it, in the lower left corner of the right atrium close to the atrio-ventricular septum, is the atrio-ventricular node (AVN)

15.9 The nodal tissue

From the AVN to the purkinje fibres

- a bundle of nodal fibres, the atrio-ventricular bundle or AV bundle, continues from the AVN
- it passes through the atrio-ventricular septa, emerges on the top of the inter-ventricular septum and immediately divides into a right and a left bundle
- these branches give rise to minute fibres spread throughout the ventricular musculature of their own sides, called purkinje fibres

The pacemaker

- the nodal musculature can generate action potentials without any external stimuli, that is, it is autoexcitable
- the number of action potentials generated in a minute varies at different parts of the nodal system
- the SAN generates the maximum number, 70-75 min–1, and so initiates and maintains the rhythmic contractile activity of the heart, which is why it is called the pacemaker
- the heart normally beats 70-75 times in a minute, an average of 72 beats min–1

15.10 The cardiac cycle

From joint diastole to ventricular systole

- to begin with, all four chambers of the heart are relaxed, a state called joint diastole
- the tricuspid and bicuspid valves are open, so blood from the pulmonary veins and the vena cava flows through the left and the right atria into the left and the right ventricles, while the semilunar valves stay closed at this stage
- the SAN then generates an action potential that makes both atria contract at the same time, the atrial systole, which raises the flow of blood into the ventricles by about 30 per cent
- the action potential is conducted to the ventricular side by the AVN and the AV bundle, from where the bundle of His transmits it through the entire ventricular musculature, making the ventricular muscles contract in the ventricular systole while the atria relax in a diastole that coincides with it

The rest of the cycle

- ventricular systole raises the ventricular pressure, which closes the tricuspid and bicuspid valves as blood attempts to flow back into the atria
- as the pressure rises further, the semilunar valves guarding the pulmonary artery on the right side and the aorta on the left side are forced open, and the blood in the ventricles flows through these vessels into the circulatory pathways
- the ventricles then relax in ventricular diastole, and the falling pressure closes the semilunar valves, which prevents backflow into the ventricles
- as the pressure declines further, the pressure of the blood emptied into the atria by the veins pushes the tricuspid and bicuspid valves open, blood again moves freely to the ventricles, the chambers return to joint diastole, and the SAN soon fires again and the sequence is repeated

What a cardiac cycle is

- this sequential event in the heart, cyclically repeated, is called the cardiac cycle
- it consists of the systole and the diastole of both the atria and the ventricles
- the heart beats 72 times per minute, so that many cardiac cycles are performed per minute
- from this it can be deduced that the duration of one cardiac cycle is 0.8 seconds

15.11 Stroke volume and cardiac output

The volumes

- during a cardiac cycle each ventricle pumps out approximately 70 mL of blood, which is called the stroke volume
- the stroke volume multiplied by the heart rate, the number of beats per minute, gives the cardiac output
- cardiac output can therefore be defined as the volume of blood pumped out by each ventricle per minute, and averages 5000 mL or 5 litres in a healthy individual
- the body can alter the stroke volume as well as the heart rate and thereby the cardiac output, which is why the cardiac output of an athlete is much higher than that of an ordinary man

The two heart sounds

- during each cardiac cycle two prominent sounds are produced, and can be easily heard through a stethoscope
- the first heart sound, lub, is associated with the closure of the tricuspid and bicuspid valves
- the second heart sound, dub, is associated with the closure of the semilunar valves
- these sounds are of clinical diagnostic significance

15.12 Electrocardiogram

What an ECG is

- the monitoring machine of a hospital television scene, which traces voltage on a screen and pips until the patient goes into cardiac arrest, is an electro-cardiograph
- this type of machine is used to obtain an electrocardiogram (ECG)
- an ECG is a graphical representation of the electrical activity of the heart during a cardiac cycle

P wave, the depolarisation of the atria

QRS complex, the depolarisation of the ventricles

T wave, the repolarisation of the ventricles

Trace, peaks lettered P to T across one cardiac cycle

Figure 15.C Diagrammatic presentation of a standard ECG

15.13 The waves of a standard ECG

Recording it

- three electrical leads, one to each wrist and one to the left ankle, connect the patient to the machine and continuously monitor the heart activity
- a detailed evaluation of the heart's function needs multiple leads on the chest region
- each peak is identified with a letter from P to T, each answering to a specific electrical activity of the heart

P, QRS and T

- the P-wave represents the electrical excitation, or depolarisation, of the atria, which leads them both to contract
- the QRS complex represents the depolarisation of the ventricles, which initiates their contraction; it starts shortly after Q and marks the beginning of the systole
- the T-wave represents the return of the ventricles from the excited to the normal state, that is repolarisation, and its end marks the end of systole
- counting the QRS complexes in a given period gives the heart beat rate; ECGs of different individuals have roughly the same shape for a given lead configuration, so any deviation points to a possible abnormality or disease, which is its clinical significance

15.14 Double circulation

The wall of an artery or a vein

- blood flows by a fixed route through the blood vessels, the arteries and veins
- each artery and vein has three layers: an inner lining of squamous endothelium, the tunica intima; a middle layer of smooth muscle and elastic fibres, the tunica media; and an outer layer of fibrous connective tissue with collagen fibres, the tunica externa
- the tunica media is comparatively thin in the veins

The two circuits

- the right ventricle pumps blood into the pulmonary artery, the left ventricle into the aorta
- the deoxygenated blood sent into the pulmonary artery passes to the lungs, from where the pulmonary veins carry oxygenated blood into the left atrium, the pulmonary circulation
- the oxygenated blood entering the aorta travels by arteries, arterioles and capillaries to the tissues, from where venules, veins and vena cava collect the deoxygenated blood and empty it into the right atrium, the systemic circulation
- the systemic circulation brings nutrients, O2 and other essentials to the tissues and takes CO2 and other harmful substances away for elimination

Pulmonary circuit, right ventricle to the lungs to the left atrium

Systemic circuit, left ventricle to the aorta to the tissues to the vena cava and the right atrium

Hepatic portal system, intestine to the liver before the systemic circulation

Coronary system, serving the cardiac musculature itself

Figure 15.D Schematic plan of blood circulation in human

Two special routes

- a unique vascular connection exists between the digestive tract and the liver, called the hepatic portal system
- the hepatic portal vein carries blood from the intestine to the liver before it is delivered to the systemic circulation
- a special coronary system of blood vessels is present in the body exclusively for the circulation of blood to and from the cardiac musculature

15.15 Regulation of cardiac activity

Myogenic, then moderated

- the normal activities of the heart are regulated intrinsically, that is auto regulated by the specialised muscles of the nodal tissue, which is why the heart is called myogenic
- a special neural centre in the medulla oblangata can moderate cardiac function through the autonomic nervous system (ANS)
- neural signals through the sympathetic nerves, a part of the ANS, can increase the rate of heart beat, the strength of ventricular contraction and thereby the cardiac output
- parasympathetic neural signals, another component of the ANS, decrease the rate of heart beat and the speed of conduction of the action potential and thereby the cardiac output, while adrenal medullary hormones can also increase the cardiac output

15.16 Disorders of the circulatory system

High blood pressure, or hypertension

- hypertension is the term for blood pressure that is higher than the normal 120/80
- in that measurement 120 mm Hg, millimetres of mercury pressure, is the systolic or pumping pressure, and 80 mm Hg is the diastolic or resting pressure
- repeated checks reading 140/90, that is 140 over 90, or higher show hypertension

Where high blood pressure leads

- high blood pressure leads to heart diseases
- it also affects vital organs such as the brain and the kidney

Coronary artery disease (CAD)

- often referred to as atherosclerosis, it affects the vessels that supply blood to the heart muscle
- it is caused by deposits of calcium, fat, cholesterol and fibrous tissues
- these deposits make the lumen of the arteries narrower

Angina

- also called angina pectoris
- it is a symptom of acute chest pain that appears when not enough oxygen is reaching the heart muscle
- it can occur in men and women of any age, but is more common among the middle-aged and the elderly
- it occurs because of conditions that affect the blood flow

Heart failure

- the state of the heart when it is not pumping blood effectively enough to meet the needs of the body
- it is sometimes called congestive heart failure, because congestion of the lungs is one of the main symptoms of this disease
- it is not the same as cardiac arrest, when the heart stops beating, nor as a heart attack, when the heart muscle is suddenly damaged by an inadequate blood supply

15.17 Summary

The fluids that carry

- vertebrates circulate blood, a fluid connective tissue, to take essential substances to the cells and to carry waste substances away from them, and lymph, the tissue fluid, is another fluid used for the transport of certain substances
- blood comprises a fluid matrix, plasma, and formed elements, which are the red blood cells or erythrocytes, the white blood cells or leucocytes and the platelets or thrombocytes
- the spaces between the cells in the tissues hold a fluid derived from blood, and this lymph is almost similar to blood except for the protein content and the formed elements

Groups, and the pump

- human blood is grouped into the A, B, AB and O systems on the presence or absence of the two surface antigens A and B on the RBCs, and grouped again on the presence or absence of another antigen, the Rhesus factor (Rh), on the surface of the RBCs
- all vertebrates and a few invertebrates have a closed circulatory system, ours consisting of a muscular pumping organ, the heart, a network of vessels and a fluid, blood
- the heart has two atria and two ventricles, and the cardiac musculature is auto-excitable
- the sino-atrial node generates the maximum number of action potentials per minute, 70-75/min, so it sets the pace of the activities of the heart and is called the pacemaker

One cycle, and its numbers

- the action potential causes the atria and then the ventricles to undergo contraction, the systole, followed by their relaxation, the diastole
- the systole forces the blood to move from the atria to the ventricles and on to the pulmonary artery and the aorta
- the sequential events in the heart, cyclically repeated, form the cardiac cycle, and a healthy person shows 72 such cycles per minute
- about 70 mL of blood is pumped out by each ventricle during a cardiac cycle, the stroke or beat volume, and the volume pumped out by each ventricle per minute is the cardiac output, equal to the product of stroke volume and heart rate, approximately 5 litres

Summary, continued

- the electrical activity of the heart can be recorded from the body surface by using an electrocardiograph, and the recording, the electrocardiogram (ECG), is of clinical importance
- we have a complete double circulation, that is two circulatory pathways, the pulmonary and the systemic
- the pulmonary circulation starts with the right ventricle pumping deoxygenated blood, which is carried to the lungs, oxygenated there and returned to the left atrium
- the systemic circulation starts with the left ventricle pumping oxygenated blood into the aorta, which carries it to all the body tissues, and the veins collect the deoxygenated blood from there and return it to the right atrium; though the heart is autoexcitable, its functions can be moderated by neural and hormonal mechanisms

15.18 Exercises

From this chapter

1 Name the components of the formed elements in the blood and mention one major function of each of them.
2 What is the importance of plasma proteins?
3 Match Column I with Column II : Column I (a) Eosinophils (b) RBC (c) AB Group (d) Platelets (e) Systole ; Column II (i) Coagulation (ii) Universal Recipient (iii) Resist Infections (iv) Contraction of Heart (v) Gas transport
4 Why do we consider blood as a connective tissue?
5 What is the difference between lymph and blood?
6 What is meant by double circulation? What is its significance?
7 Write the differences between : (a) Blood and Lymph (b) Open and Closed system of circulation (c) Systole and Diastole (d) P-wave and T-wave
8 Describe the evolutionary change in the pattern of heart among the vertebrates.
9 Why do we call our heart myogenic?
10 Sino-atrial node is called the pacemaker of our heart. Why?
11 What is the significance of atrio-ventricular node and atrio-ventricular bundle in the functioning of heart?
12 Define a cardiac cycle and the cardiac output.
13 Explain heart sounds.
14 Draw a standard ECG and explain the different segments in it.